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Exercise: Day 2

In this problem, you will work through an example of the Jarzynski equality and Crooks fluctuation theory. The system you will consider is an ideal gas of N identical particles confined in a chamber with a piston that can be moved to change the system volume, V . Since there are no interactions, the system energy is given by,

$$E = \sum_{i=1}^N \frac{1}{2} m \mathbf{v}^2.$$

1. When the piston is fixed and placed in contact with a bath at temperature $T = 1/(k_B\beta)$, the equilibrium distribution of system energies is given by,

$$P(E, T) = \frac{\Omega(E, N, V)}{Q(\beta, N, V)} e^{-\beta E}$$

Generate an expression for both $\Omega(E, N, V)$ and $Q(\beta, N, V)$.

2. Using your results from part 1, show that the distribution can be written as,

$$P(E, T) = \frac{1}{k_B T (3N/2 - 1)!} (\beta E)^{3N/2 - 1} e^{-\beta E},$$

For convenience, you should assume N is an even number.

3. Now consider changing the volume of the piston from V_0 to V_1 in the case where the container is isolated from the bath. Leveraging what you know about ideal gasses, show that the energy change for this process is, $\Delta E = E_1 - E_0 = \alpha E_0$, where $\alpha = (V_0/V_1)^{2/3} - 1$.
4. In this adiabatic process, the change in energy is equal to the work. Using the relationship, $P(W) = \langle \delta(W - \alpha E) \rangle$, show that the distribution of work values is,

$$P(W) = \frac{\beta}{|\alpha| (3N/2 - 1)!} \left(\frac{\beta W}{\alpha} \right)^{3N/2 - 1} e^{-\beta W/\alpha} \Theta(\alpha W),$$

where $\Theta(x)$ is the Heaviside step function.

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5. By computing the work to perform the reverse process, i.e., V_1 to V_0 adiabatically, show that the Crooks' work fluctuation theorem holds as,

$$\frac{P(W)}{P(-W)} = \left(\frac{V_1}{V_0} \right)^N e^{\beta W}$$

6. Make a plot of $P(W)$ versus βW for the forward and reverse processes using $N = 2$, $N = 4$, and $N = 6$ and $V_1/V_0 = 2$, and comment on the shape of the distribution.